Computational Physics I 5630 Fall 2026

Homework Assignment 1

Pseudo-Random Number Generators and Correlation Tests

In-class coding/lab. We will begin this assignment during the $\sim$ 40-minute coding/lab portion of class. The purpose is not simply to generate numbers that *look* random, but to develop simple numerical diagnostics for detecting correlations and hidden structures in a time sequence. You may use the programming language of your choice.

In-class goal. During the lab session, you should complete Parts 1–3 below and email me your code and plots at the end of class. Part 4, the autocorrelation analysis, is due Wednesday, September 2. You are encouraged to begin or finish it during class if time permits. You do *not* need to write your own plotting routines: you may use built-in routines or packages in Mathematica, MATLAB, Python, Gnuplot, or other software of your choice for the histograms and three-dimensional plots.

1. Choose a reference pseudo-random number generator.

Before implementing your own generator, identify a high-quality pseudo-random number generator (PRNG) available in the programming language or numerical library that you normally use. Find out, at least briefly, what generator or package you are using and how to specify its seed. You will use this PRNG as a reference for comparison below.

Generate a sequence of random numbers x_k uniformly distributed in the interval $[0, 1)$ and verify that your code works.

2. Implement a linear congruential generator and basic tests. (20%)

A simple pseudo-random number generator is the *linear congruential generator* (LCG),

$$X_{k+1} = (aX_k + c) \bmod m, \tag{1}$$

where X_k is an integer, m is the modulus, and a and c are integer parameters. The initial value X_0 is the seed.

Implement the LCG yourself. For this assignment, use

$$a = 63, \qquad c = 319, \qquad m = 65537.$$

Convert the integer sequence to pseudo-random numbers in $[0, 1)$ according to

$$x_k = \frac{X_k}{m}.$$

Generate a sufficiently long sequence from both your LCG and your reference PRNG. As a first check, plot a short segment of each sequence versus the sequence index k. A time-sequence plot by itself, however, may not reveal whether correlations or other structures are present.

Next, generate $N = 10{,}000$ random numbers from each generator and make a histogram for each sequence. Check whether the random numbers appear to be uniformly distributed over $[0, 1)$. Notice that a correct one-point distribution does not by itself guarantee that successive random numbers are independent.

3. Sampling points in a three-dimensional space. (40%)

As a first application, use each PRNG to sample points uniformly inside the three-dimensional unit cube. One simple way to do this is to generate three random-number sequences, each starting from a different seed:

$$\{x_1, \ldots, x_N\}, \qquad \{y_1, \ldots, y_N\}, \qquad \{z_1, \ldots, z_N\}.$$

The three numbers at the same position k define a point

$$\mathbf{r}_k = (x_k, y_k, z_k)$$

in the unit cube. Try $N = 10{,}000$ or 100,000 points. Make a three-dimensional scatter plot of the points $\mathbf{r}_k$ for the LCG and another for your reference PRNG.

Explore the plots from different viewing angles. Ask whether the sampled points appear to fill the cube uniformly, or whether you can detect any geometrical organization or correlation. Describe what you observe; there is no target plot that you are expected to reproduce.

4. Autocorrelation function. (40%; due Wednesday, September 2)

A more quantitative test is to examine correlations between random numbers generated at different positions in the sequence. For a sequence $\{x_k\}$, define the autocorrelation function at lag n by

$$A(n) = \langle x_k x_{k+n} \rangle - \langle x \rangle^2 . \tag{2}$$

Here k is the starting position and n is the separation, or lag, between two numbers in the sequence. For a sequence $x_0, \ldots, x_{N-1}$, and for a fixed lag n, average over all starting positions k for which both x_k and x_{k+n} exist; that is, $k = 0, 1, \ldots, N - n - 1$. Numerically,

$$\langle x_k x_{k+n} \rangle = \frac{1}{N-n} \sum_{k=0}^{N-n-1} x_k x_{k+n}, \tag{3}$$

with $\langle x \rangle$ obtained by averaging the random numbers in the sequence. Thus, for each value of n, you slide the pair (x_k, x_{k+n}) through the available sequence and average over the different starting positions k.

Write a routine to compute $A(n)$ for a range of lags and apply exactly the same routine to your LCG and reference PRNG. Start with relatively small lags and then extend the range of n substantially. You may also vary the total sequence length to decide whether features in $A(n)$ are robust or are finite-sample fluctuations.

Before interpreting the numerical curves, think about what $A(n)$ should look like for an ideal sequence of independent random numbers. Then compare the two generators with that expectation.

What to submit.

At the end of class, email me: (1) your code; (2) representative plots of x_k versus k for both the LCG and your reference PRNG; (3) histograms of 10,000 random numbers for both generators; and (4) the three-dimensional scatter plots. Also submit (5) your autocorrelation analysis from Part 4 by Wednesday, September 2. Part 1 carries no separate credit. Identify the reference PRNG you used and briefly explain what the different diagnostics tell you about the two generators. The emphasis is on interpreting the numerical evidence rather than reproducing any predetermined result.